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Smart Predict: C-Based Concurrent Predictive Maintenance and Emergency Machine Failure Detection System

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PROBLEM STATEMENT

- Manufacturing plants contain hundreds of machines.
- Machines generate temperature, vibration, pressure, and operating-time data.
- Abnormal behavior may lead to unexpected machine failure.
- Manual monitoring of many machines is difficult.
- Early detection is required to reduce downtime and maintenance risks.

PROPOSED SOLUTION

- Collect machine sensor readings dynamically.
- Analyze temperature and vibration against configured thresholds.
- Classify machines as **Normal** or **Emergency**.
- Generate an emergency maintenance notification.
- Store monitoring results for future reference.

TECHNOLOGIES & TECHNIQUES

- **Structures & Pointers** – manage machine information and states.
- **Dynamic Memory** – allocate memory for machines at runtime.
- **Functions & Recursion** – organize data analysis and hierarchy processing.
- **Threads & Mutexes** – process machines concurrently and protect shared data.
- **File Handling, CLI & IPC** – store history, configure thresholds, and send alerts.



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ARCHITECTURE

Sensor Data → Thread Processing → Threshold Analysis



Normal / Emergency



Alert + History File



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PROTOTYPE & RESULTS

- Thresholds are configured using command-line arguments.
- Multiple machines are monitored concurrently.
- Emergency is detected when temperature **and** vibration exceed limits.
- Emergency machine details are displayed immediately.
- A dashboard shows Normal and Emergency machine counts.



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DATA STORAGE & ADVANTAGES

- Results are stored in machine_history.txt.
- Historical records include sensor values and machine status.
- Previous monitoring sessions can be viewed later.
- Configurable thresholds make the system flexible.
- Early alerts can help reduce unexpected machine downtime.

```
hackathon x
| :
C:\Users\sound\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\sound\PycharmProjects\PythonProject\hackathon.py

CONFIGURE THRESHOLDS
Enter Temperature Threshold (C): 80
Enter Vibration Threshold (mm/s): 70

=====
          SMART MANUFACTURING PREDICTIVE MAINTENANCE
=====

Configured Thresholds
Temperature : 80.00 C
Vibration   : 70.00 mm/s

Enter number of machines: 3

-----
ENTER SENSOR DATA FOR MACHINE M001
-----

Temperature (C): 65
Vibration (mm/s): 35
Pressure (bar): 40
Operating Hours: 1000
```

```
REASON:
Temperature 90.00 > 80.00
Vibration 85.00 > 70.00

ACTION:
>>> IMMEDIATE MAINTENANCE REQUIRED <<<
=====

Data successfully saved to: machine_history.txt

=====
          MAIN MENU
=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 80
Invalid choice. Please select 1 to 5.
```

```
hackathon x
| :

-----
ENTER SENSOR DATA FOR MACHINE M002
-----

Temperature (C): 90
Vibration (mm/s): 85
Pressure (bar): 40
Operating Hours: 2450

-----
ENTER SENSOR DATA FOR MACHINE M003
-----

Temperature (C): 72
Vibration (mm/s): 45
Pressure (bar): 48
Operating Hours: 1500

=====
          EMERGENCY ALERT
=====

Machine       : M002
Temperature   : 90.00 C
Vibration     : 85.00 mm/s
Pressure      : 40.00 bar
Operating Time: 2450.00 hours
```

```
=====
          MAIN MENU
=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 70
Invalid choice. Please select 1 to 5.

=====
          MAIN MENU
=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit
```


Enter your choice: 3

FACTORY HIERARCHY

=====

```
|-- Smart Manufacturing Plant
    |-- Production Unit
        |-- Machine Group A
        |-- Machine Group B
    |-- Maintenance Unit
        |-- Sensor Processing
        |-- Emergency Alert System
```

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 65

Invalid choice. Please select 1 to 5.

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 35

Invalid choice. Please select 1 to 5.

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 40

Invalid choice. Please select 1 to 5.

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 1000

Invalid choice. Please select 1 to 5.

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 90

Invalid choice. Please select 1 to 5.

⏏ :

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 85

Invalid choice. Please select 1 to 5.

=====

MAIN MENU

=====

1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 52

Invalid choice. Please select 1 to 5.

```
=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 2450
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 72
Invalid choice. Please select 1 to 5.
```

```
=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 45
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 48
Invalid choice. Please select 1 to 5.
```

```
=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 1500
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit
```

```
Enter your choice: 1

=====
                        LIVE DASHBOARD
=====
Machine      Temperature      Vibration      Status
-----
M001         65.00             35.00          NORMAL
M002         90.00             85.00          EMERGENCY
M003         72.00             45.00          NORMAL
-----

Total Machines      : 3
Normal Machines     : 2
Emergency Machines  : 1

CONFIGURED THRESHOLDS
Temperature : 80.00 C
Vibration   : 70.00 mm/s
=====
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 2
Enter machine number: 2

=====
MACHINE DETAILS
=====
Machine ID      : M002
Machine Name    : Machine M002
Temperature     : 90.00 C
Vibration       : 85.00 mm/s
Pressure        : 40.00 bar
Operating Hours : 2450.00
Status         : EMERGENCY
=====
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 3

FACTORY HIERARCHY
=====
|-- Smart Manufacturing Plant
    |-- Production Unit
        |-- Machine Group A
        |-- Machine Group B
    |-- Maintenance Unit
        |-- Sensor Processing
        |-- Emergency Alert System
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 4

=====
HISTORICAL DATA
=====

=====
PREDICTIVE MAINTENANCE MONITORING SESSION
=====
Temperature Limit: 80.00 C
Vibration Limit   : 70.00 mm/s
=====
M001 | Temperature=65.00 | Vibration=35.00 | Pressure=40.00 | OperatingHours=1000.00 | Status=NORMAL
M002 | Temperature=90.00 | Vibration=85.00 | Pressure=40.00 | OperatingHours=2450.00 | Status=EMERGENCY
M003 | Temperature=72.00 | Vibration=45.00 | Pressure=48.00 | OperatingHours=1500.00 | Status=NORMAL
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 5

Predictive Maintenance System closed successfully.
```

FUTURE SCOPE + CONCLUSION

- Integrate real IoT sensors.
- Add machine-learning-based failure prediction.
- Develop a web or graphical dashboard.
- Successfully developed a C-based predictive maintenance prototype.
- Combines **dynamic memory, pointers, recursion, threads, mutexes, IPC, and file handling.**
- Enables early detection and emergency maintenance notification.

Title:

C-Based Predictive Maintenance System for Smart Manufacturing

A smart manufacturing plant contains hundreds of machines. Each machine produces vibration, temperature, pressure, and operating-time data. The monitoring system must detect abnormal machine behavior before failure occurs.

Develop a C-based Predictive Maintenance System that:

- Dynamically allocates memory for machines and sensor readings.
- Uses functions for data collection and analysis.
- Uses pointers to update machine states.
- Uses recursion to analyze hierarchical machine/subsystem information.
- Uses pthreads to process sensor streams concurrently.
- Uses mutexes to protect shared machine-status information.
- Stores historical data in files.
- Uses command-line arguments for threshold configuration.
- Uses IPC between sensor-processing and alert-generation modules.

Hackathon Challenge: Detect a machine whose temperature and vibration simultaneously exceed their thresholds and generate an emergency maintenance notification.

Guided By:

Dr.S.K.Saravanan

Presented By:

Teja Sri K

Soundharya S

1. Abstract

This project develops a C-based Predictive Maintenance System for monitoring machines in a smart manufacturing plant. The system collects temperature, vibration, pressure, and operating-time data and analyzes them concurrently using pthreads. Dynamic memory allocation, pointers, mutexes, recursion, file handling, command-line arguments, and IPC are used. The system detects a critical machine when both temperature and vibration exceed their threshold values and generates an emergency maintenance alert.

2. Problem Statement

In a smart manufacturing plant, machines continuously generate sensor data. If abnormal conditions are not detected at an early stage, machine failure can cause production loss and increased maintenance costs. Therefore, a system is required to continuously monitor machine sensor values and generate an emergency alert when critical conditions are detected.

3. Objective

- Monitor machine sensor data.
- Detect abnormal machine conditions.
- Process sensor data using pthreads.
- Protect shared data using mutexes.
- Store historical data in files.
- Generate an emergency alert when temperature and vibration exceed thresholds simultaneously.

4. Working

Sensor data is collected for each machine and processed concurrently using threads. The system compares temperature and vibration values with user-defined thresholds. If both values exceed their thresholds, the machine status is updated to CRITICAL. An emergency alert is then sent using IPC, and the machine information is stored in a history file.

5. System Overview

SYSTEM OVERVIEW

↓

System Features

↓

Input Parameters

↓

Machine Status Levels

↓

System Architecture Diagram

System Features

- Supports monitoring of multiple machines.
- Dynamically allocates memory for machine data.
- Collects temperature, vibration, pressure, and operating-time values.
- Uses functions for data collection and analysis.
- Uses pointers to update machine status.
- Processes sensor readings concurrently using pthreads.
- Uses mutexes for thread synchronization.
- Uses recursion for machine/subsystem hierarchy.
- Stores historical data in files.
- Accepts threshold values through command-line arguments.
- Uses IPC for alert communication.
- Generates emergency maintenance notifications.

4. Machine Status Levels

Status	Condition
NORMAL	Sensor values are within limits
WARNING	One parameter exceeds its limit
CRITICAL	Temperature and vibration exceed limits
EMERGENCY	Immediate maintenance notification generated

Concurrent Processing:

The system uses POSIX threads to process sensor readings concurrently. Each thread can analyze the data of a machine independently. This improves the performance of the system and makes it suitable for monitoring a large number of machines.

Mutex Synchronization:

Multiple threads may attempt to update machine-status information at the same time. A mutex is used to protect this shared information and prevent race conditions. The mutex ensures that only one thread can update the shared machine status at a particular time.

File Storage:

The analyzed machine information is stored in a historical data file. The file contains sensor values and the current machine status. This historical information can be used later for maintenance analysis and performance monitoring.

IPC Alert Generation:

Inter-Process Communication is used between the sensor-processing module and the alert-generation module. When a critical machine condition is detected, the sensor-processing module sends the machine information to the alert module. The alert module then generates an emergency maintenance notification.

Emergency Alert:

An emergency condition occurs when both temperature and vibration simultaneously exceed their configured threshold values. This condition may indicate a serious machine problem. The system immediately changes the machine status and generates a maintenance alert.

Hackathon Challenge

The main challenge of this project is to identify a machine that has both high temperature and high vibration simultaneously.

Condition

IF Temperature > Temperature Threshold

AND Vibration > Vibration Threshold

THEN

Machine Status = CRITICAL

Generate Emergency Maintenance Alert

Example

Assume:

Parameter	Threshold	Machine Value
Temperature	80°C	95°C
Vibration	7.0 mm/s	8.5 mm/s

Since both conditions exceed their limits, the machine is considered to be in a critical state.

Algorithm:

- Step 1: Start the program.
- Step 2: Read temperature and vibration threshold values from command-line arguments.
- Step 3: Dynamically allocate memory for machine information.
- Step 4: Collect temperature, vibration, pressure, and operating-time data.
- Step 5: Create threads for concurrent machine analysis.
- Step 6: Compare temperature with the temperature threshold.
- Step 7: Compare vibration with the vibration threshold.
- Step 8: If both temperature and vibration exceed their thresholds, mark the machine as CRITICAL.
- Step 9: Use a mutex to safely update the shared machine status.
- Step 10: Send the critical machine information to the alert module using IPC.
- Step 11: Generate an emergency maintenance notification.
- Step 12: Store the machine information in a historical data file.
- Step 13: Release dynamically allocated memory.
- Step 14: Stop the program.

Outputs:

```
hackathon x
| :
C:\Users\sound\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\sound\PycharmProjects\PythonProject\hackathon.py

CONFIGURE THRESHOLDS
Enter Temperature Threshold (C): 80
Enter Vibration Threshold (mm/s): 70

=====
SMART MANUFACTURING PREDICTIVE MAINTENANCE
=====

Configured Thresholds
Temperature : 80.00 C
Vibration   : 70.00 mm/s

Enter number of machines: 3

-----
ENTER SENSOR DATA FOR MACHINE M001
-----
Temperature (C): 65
Vibration (mm/s): 35
Pressure (bar): 40
Operating Hours: 1000
```

```
hackathon x
| :

-----
ENTER SENSOR DATA FOR MACHINE M002
-----
Temperature (C): 90
Vibration (mm/s): 85
Pressure (bar): 40
Operating Hours: 2450

-----
ENTER SENSOR DATA FOR MACHINE M003
-----
Temperature (C): 72
Vibration (mm/s): 45
Pressure (bar): 48
Operating Hours: 1500

=====
EMERGENCY ALERT
=====
Machine       : M002
Temperature   : 90.00 C
Vibration     : 85.00 mm/s
Pressure      : 40.00 bar
Operating Time: 2450.00 hours
```

```

REASON:
Temperature 90.00 > 80.00
Vibration 85.00 > 70.00

ACTION:
>>> IMMEDIATE MAINTENANCE REQUIRED <<<
=====

Data successfully saved to: machine_history.txt

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 80
Invalid choice. Please select 1 to 5.

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 70
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

```

```

Enter your choice: 3

FACTORY HIERARCHY
=====
|-- Smart Manufacturing Plant
    |-- Production Unit
        |-- Machine Group A
        |-- Machine Group B
    |-- Maintenance Unit
        |-- Sensor Processing
        |-- Emergency Alert System

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 65
Invalid choice. Please select 1 to 5.

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 35
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 40
Invalid choice. Please select 1 to 5.

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 1000
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 90
Invalid choice. Please select 1 to 5.

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 85
Invalid choice. Please select 1 to 5.

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 52
Invalid choice. Please select 1 to 5.

```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 2450
Invalid choice. Please select 1 to 5.

=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 72
Invalid choice. Please select 1 to 5.
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 45
Invalid choice. Please select 1 to 5.

=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 48
Invalid choice. Please select 1 to 5.
```

```
=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 1500
Invalid choice. Please select 1 to 5.

=====
MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit
```

```

Enter your choice: 1

=====
                        LIVE DASHBOARD
=====
Machine      Temperature      Vibration      Status
-----
M001         65.00           35.00         NORMAL
M002         90.00           85.00         EMERGENCY
M003         72.00           45.00         NORMAL
-----
Total Machines      : 3
Normal Machines     : 2
Emergency Machines  : 1

CONFIGURED THRESHOLDS
Temperature : 80.00 C
Vibration   : 70.00 mm/s
=====

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 2
Enter machine number: 2

=====
                        MACHINE DETAILS
=====
Machine ID       : M002
Machine Name     : Machine M002
Temperature      : 90.00 C
Vibration        : 85.00 mm/s
Pressure         : 40.00 bar
Operating Hours  : 2450.00
Status           : EMERGENCY
=====

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 3

FACTORY HIERARCHY
=====
|-- Smart Manufacturing Plant
    |-- Production Unit
        |-- Machine Group A
        |-- Machine Group B
    |-- Maintenance Unit
        |-- Sensor Processing
        |-- Emergency Alert System

```



```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 4

=====
                        HISTORICAL DATA
=====

=====
PREDICTIVE MAINTENANCE MONITORING SESSION
=====

Temperature Limit: 80.00 C
Vibration Limit : 70.00 mm/s
-----
M001 | Temperature=65.00 | Vibration=35.00 | Pressure=40.00 | OperatingHours=1000.00 | Status=NORMAL
M002 | Temperature=90.00 | Vibration=85.00 | Pressure=40.00 | OperatingHours=2450.00 | Status=EMERGENCY
M003 | Temperature=72.00 | Vibration=45.00 | Pressure=48.00 | OperatingHours=1500.00 | Status=NORMAL

```

```

=====
                        MAIN MENU
=====
1. View Live Dashboard
2. View Machine Details
3. View Factory Hierarchy
4. View Historical Data
5. Exit

Enter your choice: 5

Predictive Maintenance System closed successfully.

```

Result

Use the short result paragraph.

Advantages

- Early detection of machine problems.
- Reduces unexpected failures.
- Improves machine reliability.
- Supports concurrent processing.
- Generates emergency alerts.

Conclusion

Add your short conclusion.

Future Enhancement (optional)

- Real IoT sensor integration.
- Machine learning-based prediction.
- Mobile or email alerts.

References:

1. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, 2nd Edition, Prentice Hall.
2. [GNU Compiler Collection \(GCC\) Documentation](#)
3. [POSIX Threads \(pthreads\) Documentation – The Open Group](#)
4. [Linux Manual Pages \(pthread, mutex, pipe and file handling\)](#)
5. [IBM – Predictive Maintenance Overview](#)
6. [GeeksforGeeks – C Programming and System Programming Concepts](#)